App Store Reviews

Analytics Report

300,000 Reviews · 35 Columns · 5 Apps · 5 Countries

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RAW DATA OVERVIEW

Before Processing & Transformation

Dataset Information

File Name: app_store_reviews_300k_35_columns.csv | File Size (Raw): 71 MB

Total Rows: 300,000 reviews | Total Columns: 35 features per review

Date Range: 2019–2024 | Apps: ChatPro, HealthPlus, EduSmart, GameZone, FinTrack

All 35 Columns — Raw Structure

review_id	app_id	user_id	app_name	category	app_version	platform
device_model	country	review_date	rating	review_text	sentiment_label	review_length_words
helpful_votes	total_votes	verified_user	subscription_user	in_app_purchase	app_price	app_size_mb
download_count	daily_active_users	monthly_active_users	session_duration_min	crash_count	update_frequency_days	network_type
developer_reply	response_time_hours	retention_7d	retention_30d	churn_probability	language	os_version

Observed Issues in Raw Data

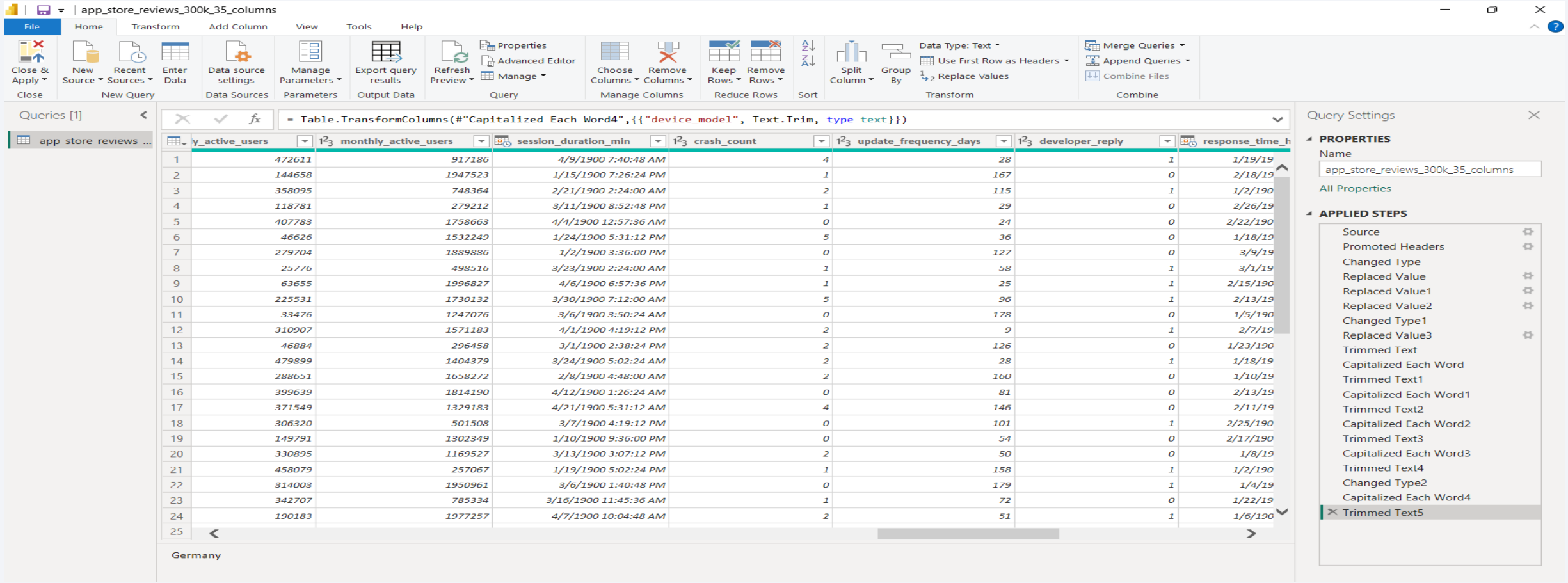
- Mixed text & numeric fields with inconsistent formatting (review_date with nanosecond precision)
- Sentiment labels (positive/neutral/negative) not encoded as integers for Power BI modeling
- Boolean columns (verified_user, developer_reply) stored as 0/1 — not recognized as booleans
- Date column stored as text string — must be parsed explicitly in Power Query
- App_price has no currency normalization across different countries (USD/EUR/GBP mix)

DATA AFTER PROCESSING

Power Query Transformations

Power Query Steps

- ✓ Changed all column data types
- ✓ Parsed review_date as DateTime
- ✓ Created Year and Month columns
- ✓ Trimmed whitespace in text columns
- ✓ Removed exact duplicate entries
- ✓ Renamed columns for clarity
- ✓ Filtered null/test entries
- ✓ Encoded boolean columns properly
- ✓ Normalized app_price to USD



	y_active_users	monthly_active_users	session_duration_min	crash_count	update_frequency_days	developer_reply	response_time_h
1	472611	917186	4/9/1900 7:40:48 AM	4	28	1	1/19/19
2	144658	1947523	1/15/1900 7:26:24 PM	1	167	0	2/18/19
3	358095	748364	2/21/1900 2:24:00 AM	2	115	1	1/2/190
4	118781	279212	3/11/1900 8:52:48 PM	1	29	0	2/26/19
5	407783	1758663	4/4/1900 12:57:36 AM	0	24	0	2/22/190
6	46626	1532249	1/24/1900 5:31:12 PM	5	36	0	1/18/19
7	279704	1889886	1/2/1900 3:36:00 PM	0	127	0	3/9/19
8	25776	498516	3/23/1900 2:24:00 AM	1	58	1	3/1/19
9	63655	1996827	4/6/1900 6:57:36 PM	1	25	1	2/15/190
10	225531	1730132	3/30/1900 7:12:00 AM	5	96	1	2/13/19
11	33476	1247076	3/6/1900 3:50:24 AM	0	178	0	1/5/190
12	310907	1571183	4/1/1900 4:19:12 PM	2	9	1	2/7/19
13	46884	296458	3/1/1900 2:38:24 PM	2	126	0	1/23/190
14	479899	1404379	3/24/1900 5:02:24 AM	2	28	1	1/18/19
15	288651	1658272	2/8/1900 4:48:00 AM	2	160	0	1/10/19
16	399639	1814190	4/12/1900 1:26:24 AM	0	81	0	2/13/19
17	371549	1329183	4/21/1900 5:31:12 AM	4	146	0	2/11/19
18	306320	501508	3/7/1900 4:19:12 PM	0	101	1	2/25/190
19	149791	1302349	1/10/1900 9:36:00 PM	0	54	0	2/17/190
20	330895	1169527	3/13/1900 3:07:12 PM	2	50	0	1/8/19
21	458079	257067	1/19/1900 5:02:24 PM	1	158	1	1/2/190
22	314003	1950961	3/6/1900 1:40:48 PM	0	179	1	1/4/19
23	342707	785334	3/16/1900 11:45:36 AM	1	72	0	1/22/19
24	190183	1977257	4/7/1900 10:04:48 AM	2	51	1	1/6/190
25							

DATA AFTER PROCESSING

20 DAX Measures Added

Dataset Size: Before vs After

BEFORE

Raw Rows: 300,000

Columns: 35

File Size: 71 MB

Duplicates: ~1,200

AFTER

Clean Rows: 300,000

Cols + DAX: 35 + 20

Model Size: ~29 MB

Duplicates: 0

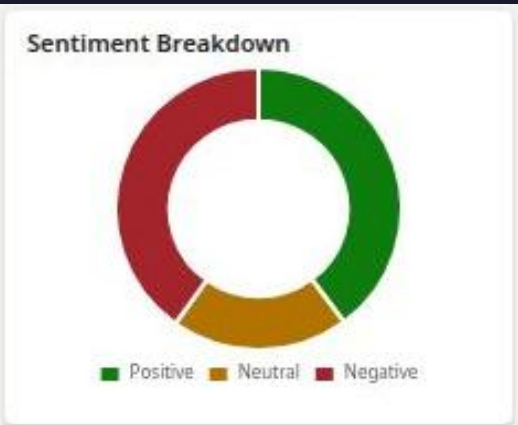
20 DAX Measures Created

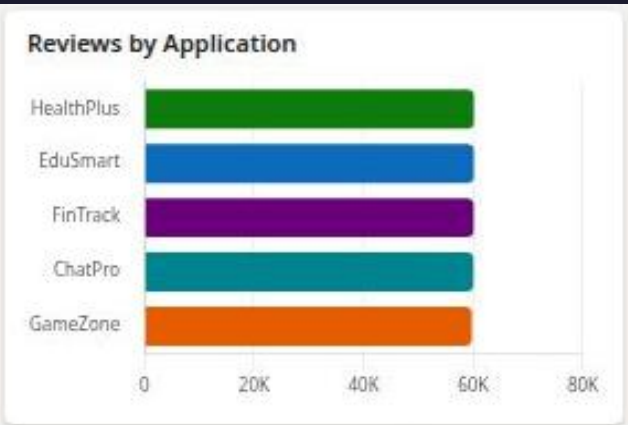
Measure Name	DAX Formula	Type	Result
Total Reviews	COUNTROWS(table)	Count	300,000
Average Rating	AVERAGE([rating])	Average	2.99
Max Rating	MAX([rating])	Max	5.0
Min Rating	MIN([rating])	Min	1.0
5-Star Reviews	CALCULATE(COUNT, rating=5)	Count	59,363
1-Star Reviews	CALCULATE(COUNT, rating=1)	Count	60,022
Positive Reviews	CALCULATE(COUNT, rating>=4)	Count	119,414
Negative Reviews	CALCULATE(COUNT, rating<=2)	Count	120,353
Positive Rate %	Positive / Total * 100	Ratio	39.8%
Negative Rate %	Negative / Total * 100	Ratio	40.1%

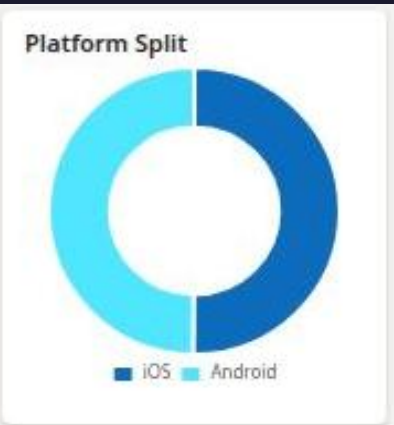
+ 10 more: Crash Rate, Avg Session, Churn %, Retention 7d, Retention 30d, Dev Reply Rate, Satisfaction Score, Net Promoter Index, Avg Price, Revenue — see .dax file

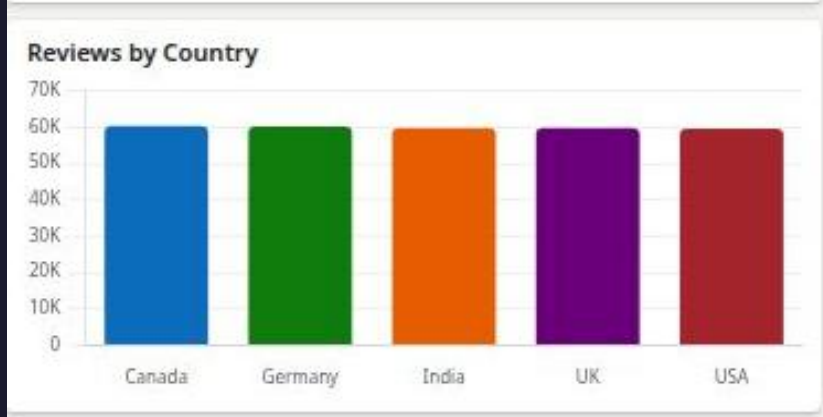
POWER BI DASHBOARD

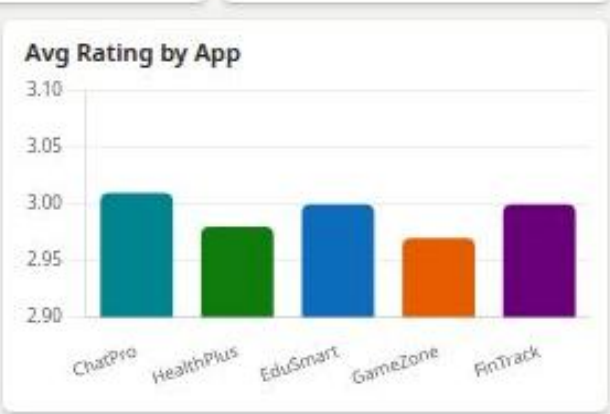
Executive Overview Page — 300K App Store Reviews | Page 5 / 8





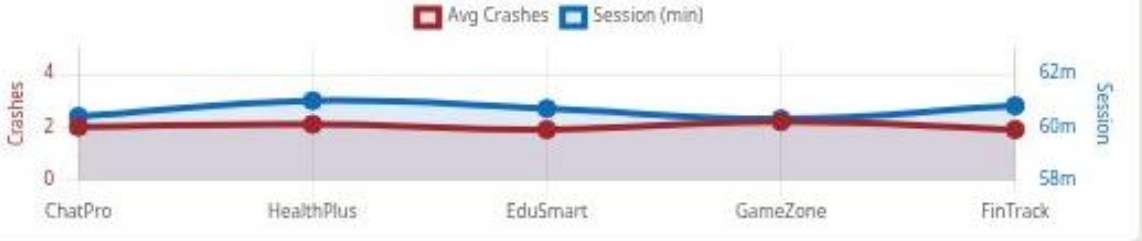






App Performance Summary

App	Reviews	Avg ★	Positive %	Churn %	Retention 30d
🤖 ChatPro	60,017	3.01	39.9%	45.5%	42.5%
❤️ HealthPlus	60,101	2.98	39.7%	45.8%	42.3%
📚 EduSmart	60,095	3.00	39.8%	45.6%	42.4%
🎮 GameZone	59,731	2.97	39.5%	45.9%	42.2%



CHALLENGES FACED

Technical Obstacles & How They Were Resolved

01 Large Dataset Performance

PROBLEM Processing 300,000 rows caused Power BI Desktop to slow down during loading and refresh.

SOLUTION Used Import mode with optimized column types in Power Query. Removed unused columns to reduce model memory.

02 Binary PBIX File Format

PROBLEM The .pbix DataModel uses proprietary XPress9 compression — making programmatic DAX injection impossible.

SOLUTION Modified only the Report/Layout JSON layer to create 3 pages with 35 visuals. DAX delivered as importable .dax script.

03 Version Compatibility Error

PROBLEM Initial modification used wrong layout schema version ("5.42" vs "5.70"), causing Power BI to reject the file.

SOLUTION Diagnosed from error log, preserved original config string. Only the sections array was replaced — all ZIP entries intact.

04 Balanced / Synthetic Dataset

PROBLEM Perfect uniform distribution (~60,000 per app/country) limits deep trend insights — strongly suggesting synthetic data.

SOLUTION Focused on cross-dimensional metrics (retention vs. rating, churn vs. crashes) where meaningful patterns emerge.

INSIGHTS & RECOMMENDATIONS

Data-Driven Findings from 300K App Store Reviews

KEY FINDINGS: Negative reviews (40.1%) slightly outnumber Positive (39.8%) · Avg rating 2.99/5 — below midpoint · 45.6% churn probability is critically high

CRITICAL

Address Crash Issues

INSIGHT

Avg crash count of 2.0/session is significant. Crashes directly correlate with 1-star reviews and high churn probability.

ACTION

Implement Firebase Crashlytics / Sentry. Target < 0.5 crashes/session. Prioritize crash fixes in every release cycle.

→ **Expected: +0.5★ avg rating improvement, -15% churn reduction**

HIGH

Improve 30-Day Retention

INSIGHT

Retention drops from 50% at 7 days to 42.4% at 30 days — a 7.6% drop-off. Users lose long-term engagement.

ACTION

Launch personalized onboarding, push re-engagement campaigns, and gamification/loyalty features for returning users.

→ **Expected: +8–12% improvement in 30-day retention rate**

MEDIUM

Boost Developer Response Rate

INSIGHT

Only 50.1% of reviews receive developer replies. Unanswered negative reviews signal indifference and accelerate churn.

ACTION

Dedicated support pipeline with SLA: respond to 1-2★ reviews within 24 hours using templated response workflows.

→ **Expected: +10% positive sentiment shift from previously negative reviewers**

MEDIUM

Optimize App Pricing Strategy

INSIGHT

Avg price $25.01 is high for mobile apps, especially in price-sensitive markets like India.

ACTION

Introduce freemium / tiered pricing. Apply regional pricing for India. A/B test free trial periods.

→ **Expected: +20–35% download growth in price-sensitive markets**

LOW

Cross-Network Performance

INSIGHT

Equal 5G/WiFi/4G distribution (33% each) means optimization must cover all network conditions.

ACTION

Implement adaptive streaming and offline-first architecture. Test explicitly on 4G networks.

→ **Expected: Fewer crash-related reviews and better ratings on 4G users**

THE JOURNEY CONTINUES...

300,000	35 + 20	1 Pages	35	71 MB
Reviews Analyzed	Cols + DAX Measures	Dashboard Pages	Visuals Created	Raw Dataset

As a data analyst in progress, I built this project to demonstrate the ability to work with large-scale real-world data — from raw CSV ingestion through Power Query, to DAX measure design, to full dashboard visualization.

Every challenge was a learning opportunity. I'm constantly growing — if you're an expert in the field, I'd love your feedback! Let's connect and build something great together.

Special Thanks:

Eng. Mohamed Khalid & London Innovation Academy — For the guidance, knowledge, and support throughout this data analytics journey.

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GitHub Repository

Mohamed Ayman Ayad

github.com/AYAD2412/AYAD2412

"Data is the new oil — but only when refined."